BIO 005 · WEEK 1 · SILVERTHORN CHAPTERS 1 AND 6

Everything these two chapters ask you to be able to do.

Each competency is written out in full, with two brain dump prompts underneath it. Pick one prompt per competency and do that one on your note sheet. You do not need to open any other list.

SILVERTHORN CHAPTER 1

Introduction to physiology

What physiology asks, homeostasis, mass balance, the reflex pathway, and the two kinds of feedback.

1 Physiology and levels of function (LECTURE)

YOU SHOULD BE ABLE TO

Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

WORKED EXAMPLE, SO YOU CAN SEE WHAT A FINISHED ONE LOOKS LIKE

Carrying oxygen, one case all the way up the ladder

Do not stop at naming the levels. Pick one thing your body does and follow it up, so each rung is the same story at a bigger scale.

MOLECULE	Hemoglobin. Four subunits, each one binds a molecule of oxygen.
CELL	The red blood cell. Packed with hemoglobin, and it drops its nucleus to make room for more.
TISSUE	Blood. Red cells suspended in plasma, moving as one fluid tissue.
ORGAN	The lung loads the oxygen on. The heart pushes the loaded blood out.
ORGAN SYSTEM	Respiratory and cardiovascular, working as one delivery line.
ORGANISM	You climb the stairs without stopping.

Now break one rung and follow the damage up

Change one amino acid at the bottom and every level above it changes too. That is how you know the levels are real and not just labels.

MOLECULE	In sickle hemoglobin, valine sits at position 6 of the beta chain where glutamic acid should be. Once it gives up its oxygen, the molecules stick to each other and build long polymers.
CELL	Those polymers push the red cell out of shape into a stiff sickle. It no longer bends to get through a capillary.
TISSUE	Sickled cells are destroyed far faster than the marrow can replace them, so the blood carries less oxygen. The stiff ones also jam in small vessels and stop flow.
ORGAN	Downstream tissue is starved of oxygen. That is what a pain crisis is, and over years it damages the spleen, the kidneys, the brain.
ORGAN SYSTEM	The delivery line runs short on both counts, less oxygen carried and blocked routes to deliver it through.
ORGANISM	Fatigue, breathlessness, episodes of severe pain.

Your turn

Do not use oxygen delivery. Pick your own process and build both halves for it, the ladder up and the break. Anything works as long as you can follow it through all six levels: contracting a muscle, digesting a meal, clotting a cut, hearing a sound, filtering blood in the kidney. Choose the one you are most curious about, because you will meet it again later in the term.

PICK ONE AND DO THAT ONE

- [ ] **A** Build your own ladder. Pick a process, not oxygen delivery, and draw it at all six levels, molecule up to organism. One small box per level, labeled. Then break one rung and write, level by level, what stops working above it.

- [ ] **B** Pick your own process, not oxygen delivery. Draw the same thing twice. First as anatomy sees it, the structures and their parts. Then as physiology sees it, the same thing working, with arrows for what moves and what changes. Under each drawing, one line naming the question it answers.

2 Structure and function relationship LECTURE LAB

YOU SHOULD BE ABLE TO

Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one structure from this week and label the single feature that makes its job possible. Then sketch what would stop working if that feature changed.
- [ ] **B** List three structures from this week. Beside each, write the one property its function depends on, in six words or fewer.

3 Homeostasis defined LECTURE

YOU SHOULD BE ABLE TO

Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

PICK ONE AND DO THAT ONE

- [ ] **A** Write the definition of homeostasis, then underline the three words doing the real work. Beside each underlined word, one line on why it matters.
- [ ] **B** Draw two graphs side by side, a value in steady state and a value at equilibrium. On each, label what is still moving and what it costs.

4 Feedback loop components LECTURE LAB

YOU SHOULD BE ABLE TO

Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw the five component reflex pathway top down and label all five plus the feedback arrow. Labels only, no sentences.
- [ ] **B** Fill in the five components for thermoregulation on a cold day. Then mark which component you would suspect first if the patient never shivered.

5 Negative and positive feedback LECTURE

YOU SHOULD BE ABLE TO

Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one negative feedback loop and one positive feedback loop as arrow diagrams. On each, mark what makes it stop.
- [ ] **B** Draw clotting as a positive feedback loop, then show what ends it. Underneath, write one line on why the body can afford to use positive feedback here.

6 Feedforward control and acclimatization LECTURE

YOU SHOULD BE ABLE TO

Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a timeline of eating a meal. Mark where feedforward starts and where feedback takes over.
- [ ] **B** Draw a setpoint as a horizontal line. Show what fever does to it, and mark at each stage what the patient feels.

7 Local and reflex control pathways LECTURE

YOU SHOULD BE ABLE TO

Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a local control example and a reflex control example side by side. On each, mark how far the signal travels.
- [ ] **B** Standing up suddenly: draw the reflex, label all five components, and write one line on why local control could not have handled it.

8 Mass balance LECTURE LAB

YOU SHOULD BE ABLE TO

Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

PICK ONE AND DO THAT ONE

- [ ] **A** Write the mass balance equation and label every term. Then draw a body outline with arrows in and out for sodium, labeling which arrow is which term.
- [ ] **B** Work a mass balance for one substance across one day. Then change a single term and show the new result underneath.

9 Body fluid compartments LECTURE LAB

YOU SHOULD BE ABLE TO

State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw the fluid compartments top down with the fractions. Use one border style for inside cells and a different one for outside.
- [ ] **B** On a compartment drawing, mark where a blood sample is taken and which compartment you are actually drawing a conclusion about.

10 Compartment separation and clinical volume shifts LECTURE

YOU SHOULD BE ABLE TO

Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a patient with swollen ankles and a low blood pressure. Use arrows to show where the water went.
- [ ] **B** Draw three scenarios: water gained in the vessels, water lost from the body, water moved into tissue. Predict blood pressure under each.

USED ALL TERM, STARTING THIS WEEK

Quantitative and lab skills

These are not a chapter. They are the skills every lab this term leans on, and this is the week you set them up.

11 Units and unit conversion LECTURE LAB

YOU SHOULD BE ABLE TO

Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

PICK ONE AND DO THAT ONE

- [ ] **A** Convert three values from this week. Show every step and carry the units through each line.
- [ ] **B** Take one equation from this week and show that the units on the left match the units on the right.

12 Graphing and data interpretation LECTURE LAB

YOU SHOULD BE ABLE TO

Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

PICK ONE AND DO THAT ONE

- [ ] **A** Plot the mission patient's sodium across the three days. Label both axes with units and mark the reference range on the graph.
- [ ] **B** Draw a saturation curve. Label the plateau, then write one line on what the plateau means physiologically.

13 Experimental design and controls LECTURE LAB

YOU SHOULD BE ABLE TO

Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

PICK ONE AND DO THAT ONE

- [ ] **A** Sketch an experiment as three labeled boxes: what you changed, what you measured, and your control.
- [ ] **B** Take a claim from this week and design the control group that would test it. Draw both groups and label the one difference between them.

14 Measurement error and variability LECTURE LAB

YOU SHOULD BE ABLE TO

Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw two sets of repeated measurements, one with high variability and one with low. Mark which difference between readings is real.
- [ ] **B** List three sources of variation in a single measurement. Beside each, one way to reduce it.

SILVERTHORN CHAPTER 6

How signals travel

The part of chapter 6 Mission 1 covers: what carries a signal, where its receptor sits, how a small signal becomes a big response, and how it gets shut off.

15 Signal types and range LECTURE

YOU SHOULD BE ABLE TO

Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw one target cell in the middle of the box. Draw an arrow in from each of the five signal types, autocrine, paracrine, neurotransmitter, neurohormone, hormone. Label each arrow with how far it travelled and what carried it.
- [ ] **B** Make a five row table. Signal type, where it is released, how far it goes, what it travels through. Fill it from memory, then check one row against the book.

16 Receptor location and ligand solubility LECTURE

YOU SHOULD BE ABLE TO

Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a cell with one surface receptor and one intracellular receptor. Show the path a lipid soluble signal takes and the path a water soluble one takes. Mark which response is faster and which lasts longer.
- [ ] **B** Write the two questions you would ask about any new signal molecule to predict where its receptor sits. Then answer both for a steroid and for a peptide hormone.

17 Signal amplification LECTURE

YOU SHOULD BE ABLE TO

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a three step cascade as a widening funnel. Put a number on each step and multiply them down the side to get the total amplification.
- [ ] **B** Start from one ligand and write the count at each step of a cascade until you reach the response. One line per step, numbers only, no sentences.

18 Signal termination LECTURE

YOU SHOULD BE ABLE TO

Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

PICK ONE AND DO THAT ONE

- [ ] **A** Draw a signal at a receptor, then draw the three ways it gets shut off. Label each one. Beside the drawing, write what goes wrong if that mechanism fails.

- [ ] **B** Name the three termination mechanisms. For each, write one clinical consequence of it failing, in one line.

NOT IN THESE CHAPTERS

Water and solution properties, pH and buffers, protein structure, enzymes, and ATP are Silverthorn chapter 2. They are Week 2, not this week. If you saw them on an earlier version of the week 1 list, that is why they are not here.

The rest of chapter 6, G protein coupled receptors, second messengers, receptor modulation and dose response, comes later in the term with the rest of cell signaling.